VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 10 | JEE 2022

Date: 27/06/2021 **Maximum Marks: 300**

Timing: 04:00 PM - 09:00 PM **Duration: 3.0 Hrs**

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : _

Roll Number : _

OMR Bar Code Number : _

Candidate's Signature : _ _ _ _ _ _ _ _ _ _ Invigilator's Signature _ _ _ _ _ _ _ _

PART I : PHYSICS 100 MARKS

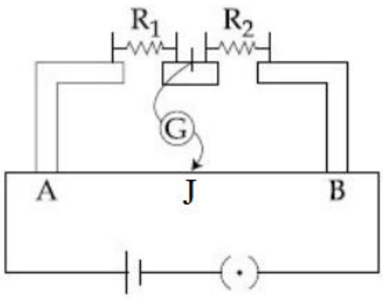
SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Three particles carrying charge Q each are fixed at the vertices of an equilateral triangle of side length a. The net electrostatic force acting on any one of the particles is:

(A) $\frac{\sqrt{3}Q^2}{8\pi\varepsilon_0 a^2}$ **(B)** $\frac{\sqrt{3}Q^2}{4\pi\varepsilon_0 a^2}$ **(C)** $\frac{Q^2}{4\pi\varepsilon_0 a^2}$ **(D)** $\frac{Q^2}{2\pi\varepsilon_0 a^2}$

2. In the metre bridge circuit shown, AB is a uniform wire of length 100 cm, and two resistances R_1 and R_2 are connected in the gaps as shown. If the current through AJ is zero for length AJ = 30 cm, the ratio $\frac{R_1}{R_2}$ is:



(A) $\frac{7}{3}$ **(B)** $\frac{3}{7}$ **(C)** $\frac{3}{10}$ **(D)** $\frac{10}{3}$

3. Two batteries E_1 and E_2 of emf 12 V each and internal resistance $1\,\Omega$ each are connected in series across a $10\,\Omega$ resistance, such that the positive terminal of E_1 is connected to the negative terminal of E_2. The potential difference (in Volts) across the terminals of E_1 becomes:

(A) 6 **(B)** 9 **(C)** 10 **(D)** 14

4. A potential difference is applied across a metallic conductor. Consider the following statements:

(I) at any instant, the velocity of all electrons in the conductor is approximately equal to the drift velocity

(II) the number of electrons passing through a cross-section of the conductor in a given time interval is proportional to the drift velocity

Which of these options is correct?

(A) Both (I) and (II) are correct **(B)** (I) is correct and (II) is incorrect

(C) (II) is correct and (I) is incorrect **(D)** Both (I) and (II) are incorrect

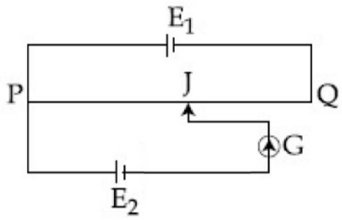
5. A non-conducting sphere of radius R is charged such that the region $r < \frac{R}{2}$ (r denotes the distance from the centre of the sphere) has uniform charge density 2ρ, and the region $\frac{R}{2} \le r \le R$ has uniform charge density ρ. The electric potential at the surface of the sphere is:

(A) $\frac{5}{8}\frac{\rho R^2}{\varepsilon_0}$ **(B)** $\frac{3}{8}\frac{\rho R^2}{\varepsilon_0}$ **(C)** $\frac{5}{4}\frac{\rho R^2}{\varepsilon_0}$ **(D)** $\frac{3}{4}\frac{\rho R^2}{\varepsilon_0}$

6. Two infinitely long wires are fixed in the X-Y plane along the lines $x=-a$ and $x=a$. The wires are charged uniformly with charge per unit length λ and $-\lambda$ respectively. The electric field at the point $(2a,0)$ is:

(A) $\frac{\lambda}{6\pi\varepsilon_0 a}\hat{i}$ **(B)** $\frac{\lambda}{3\pi\varepsilon_0 a}\hat{i}$ **(C)** $\frac{\lambda}{6\pi\varepsilon_0 a}(-\hat{i})$ **(D)** $\frac{\lambda}{3\pi\varepsilon_0 a}(-\hat{i})$

7. In the circuit shown, the potentiometer wire PQ has length 120 cm and total resistance $10\ \Omega$. The main cell E_1 has emf 24 V and internal resistance $2\ \Omega$. The jockey J is moved on the potentiometer wire PQ to find the balancing point. If the current through the galvanometer is zero for the length PJ = 70 cm, the emf of the cell E_2 is:



(A) 11.7 V **(B)** 12.4 V **(C)** 12.9 V **(D)** 13.5 V

8. A very long insulating cylinder of radius R is charged uniformly with charge per unit volume ρ. Let $E(r)$ denote the magnitude of electric field at any point that lies at a distance r from the axis of the cylinder. If for some $a<R$, and some $b>R$, $E(a)=E(b)$, then:

(A) $ab=R^2$ **(B)** $a^2b=R^3$ **(C)** $ab^2=R^3$ **(D)** $a^3b=R^4$

9. In the presence of a uniform electric field $\vec{E}$, a particle carrying charge 0.002 C is taken from the origin O to the point $A(1,0)$ along a straight line path. Then, the particle is taken from A to the point $B(1,-1)$, also along a straight line path. The coordinates are in metres. If the work done on the charge by the electric field $\vec{E}$ in the journeys $O\to A$ and $A\to B$ is respectively 4 J and 3 J, the electric field $\vec{E}$ is:

(A) $2500\left(\frac{4\hat{i}-3\hat{j}}{5}\right)$ V/m **(B)** $2500\left(\frac{-4\hat{i}+3\hat{j}}{5}\right)$ V/m

(C) $500\left(\frac{4\hat{i}-3\hat{j}}{5}\right)$ V/m **(D)** $500\left(\frac{-4\hat{i}+3\hat{j}}{5}\right)$ V/m

10. When two resistances R_1 and R_2 (such that $R_1<R_2$) are connected in series across an ideal battery, the rate of heat dissipation is H_1. When the resistances are connected in parallel across the same battery, the rate of heat dissipation is H_2. If the ratio $H_1:H_2=4:25$, the ratio $R_1:R_2$ is:

(A) 2:3 **(B)** 1:3 **(C)** 1:2 **(D)** 1:4

11. A small electric dipole with dipole moment p is fixed at the origin with its dipole moment vector pointing in the +X direction. The electric field due to the dipole at the point (a,a) is:

(A) $\left(\frac{\sqrt{5}}{16}\right)\left(\frac{p}{\pi\varepsilon_0 a^3}\right)\left(\frac{-\hat{i}+3\hat{j}}{\sqrt{10}}\right)$ **(B)** $\left(\frac{\sqrt{5}}{16}\right)\left(\frac{p}{\pi\varepsilon_0 a^3}\right)\left(\frac{\hat{i}+3\hat{j}}{\sqrt{10}}\right)$

(C) $\left(\frac{\sqrt{5}}{16\sqrt{2}}\right)\left(\frac{p}{\pi\varepsilon_0 a^3}\right)\left(\frac{\hat{i}+2\hat{j}}{\sqrt{5}}\right)$ **(D)** $\left(\frac{\sqrt{5}}{16\sqrt{2}}\right)\left(\frac{p}{\pi\varepsilon_0 a^3}\right)\left(\frac{-\hat{i}+2\hat{j}}{\sqrt{5}}\right)$

12. A galvanometer of resistance $10\,\Omega$ breaks down if the current through it exceeds 0.1 A. To convert it into an ammeter that can measure current up to 10 A, a shunt resistance is connected in parallel with it. The net resistance (in Ohms) of the galvanometer now becomes:

(A) 0.01 **(B)** 0.05 **(C)** 0.1 **(D)** 0.2

13. A small dipole is placed on the axis of a circular ring, with its dipole moment vector pointing along the axis, away from the centre of the ring. The ring is charged uniformly over its length. The dipole is rotated by 180° such that now its dipole moment vector points directly towards the centre of the ring. Let the work done in the process be denoted by *W*. Which of these options is correct?

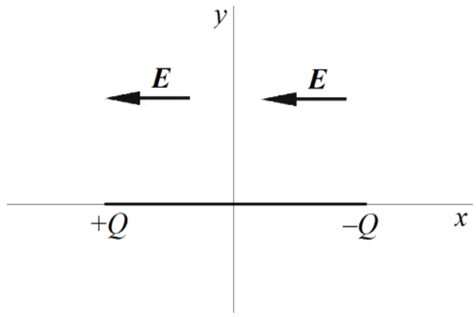
(A) $W > 0$ if the ring is positively charged and $W < 0$ if the ring is negatively charged

(B) $W < 0$ if the ring is positively charged and $W > 0$ if the ring is negatively charged

(C) $W > 0$

(D) $W < 0$

14. A massless non-conducting rod of length *L* lies along the X-axis with its centre at the origin. The two ends of the rod carry point charges +*Q* and –*Q* as shown. A uniform electric field of magnitude *E* exists in the –X direction. The work done in slowly rotating the rod about the origin counter-clockwise by 60° is:

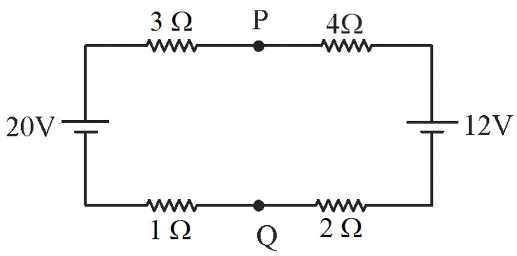


(A) $\frac{QEL}{2}$ **(B)** QEL **(C)** $\frac{\sqrt{3}QEL}{2}$ **(D)** $\sqrt{3}QEL$

15. A particle of mass *m* and carrying charge +*q* moves in a gravity-free region with a uniform electric field $\vec{E}$ present. At *t* = 0, the particle has velocity *v*. At $t = t_0$, the particle has velocity $\sqrt{3}v$ in a direction perpendicular to its direction at *t* = 0. The magnitude of the electric field $\vec{E}$ is:

(A) $\frac{\sqrt{3}mv}{qt_0}$ **(B)** $\frac{2mv}{qt_0}$ **(C)** $\frac{mv}{qt_0}$ **(D)** $\frac{2\sqrt{3}mv}{qt_0}$

16. Both batteries in the given circuit are ideal. If an ideal voltmeter is connected between the points P and Q, it shows a reading:



(A) 21.6 V **(B)** 16.8 V **(C)** 14.4 V **(D)** zero

17. A non-ideal battery of internal resistance $1\ \Omega$ is connected across a wire of resistance $20\ \Omega$. The current through the battery is I_0. Now, the wire is cut into 4 equally long pieces, and the pieces are connected in parallel across the same battery. The current through the battery is now:

(A) $\frac{28}{5}I_0$ **(B)** $\frac{14}{5}I_0$ **(C)** $\frac{28}{3}I_0$ **(D)** $\frac{14}{3}I_0$

18. Two thin conducting spherical shells of radius R and $2R$ are fixed concentrically and given total charge Q_1 and Q_2 respectively. Let r denote the distance from their common centre. Consider the following statements:

(I) the potential difference between the shells depends on Q_1, but not on Q_2

(II) the electric field in the region $r > 2R$ is proportional to $Q_1 + Q_2$

Which of these options is correct?

(A) Both (I) and (II) are correct **(B)** (I) is correct and (II) is incorrect

(C) (II) is correct and (I) is incorrect **(D)** Both (I) and (II) are incorrect

19. Consider a hemispherical surface with its centre at the origin, and its circular base entirely in the X-Y plane. A uniform electric field $\vec{E}$ exists. Then, the net electric flux through the hemispherical surface depends on:

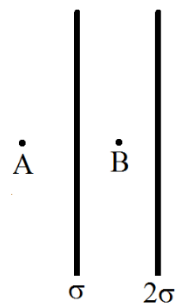
(A) only the X and Y components of $\vec{E}$

(B) only the Z component of $\vec{E}$

(C) the X, Y and Z components of $\vec{E}$

(D) only the magnitude of $\vec{E}$, and is independent of its individual components

20. Two large non-conducting plates are fixed parallel to each other with a very small distance between them. The plates are charged uniformly with charge per unit area σ and 2σ as shown. If the electric field at the points A and B is $\overrightarrow{E_A}$ and $\overrightarrow{E_B}$ respectively, then the correct relation is:



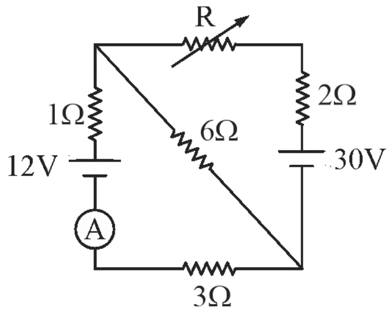
(A) $\overrightarrow{E_A} = -\frac{3}{2}\overrightarrow{E_B}$ **(B)** $\overrightarrow{E_A} = \frac{3}{2}\overrightarrow{E_B}$ **(C)** $\overrightarrow{E_A} = -3\overrightarrow{E_B}$ **(D)** $\overrightarrow{E_A} = 3\overrightarrow{E_B}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*.** **(Example: 6, 81, 1.50, 3.25, 0.08)**

21. Both batteries and the ammeter in the given circuit are ideal, and *R* is a variable resistance. The value of *R* (in Ohms) for which the ammeter shows zero reading is ___________.

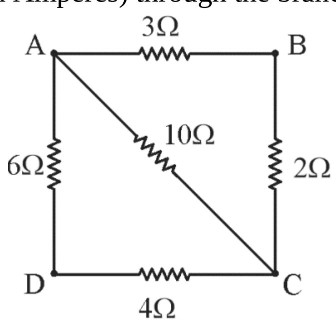


22. A potential difference 100 V is applied across a metal wire of length 1 m. If the resistivity of the metal is $10^{-7}\ \Omega\text{m}$, and the current density in the wire is $10^{N}\ \text{A/m}^2$, then *N* is __________.

23. Two particles, carrying charge +*Q* each, are fixed at the points $(-a, 0)$ and $(a, 0)$. A third particle, of mass *m* and carrying charge –*Q*, is released from rest from the point $(0, \sqrt{3}\,a)$. If in its subsequent motion, the maximum velocity attained by this particle is $\left(\frac{1}{N}\left(\frac{Q^2}{\pi \varepsilon_0 a m}\right)\right)^{1/2}$, then *N* is ____________.

24. A simple pendulum with a bob of mass 100 g has time period 4 s. If the bob is given charge 0.001 C and a vertically upward electric field of magnitude 360 V/m is switched on, the time period (in seconds) of the pendulum becomes _____________. $\left(g = 10\ \text{m/s}^2\right)$

25. In the given network, a battery of emf 15 V and internal resistance $1.4\ \Omega$ is connected directly between the points B and C. The current (in Amperes) through the branch BC is __________.



SPACE FOR ROUGH WORK

PART II : CHEMISTRY 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

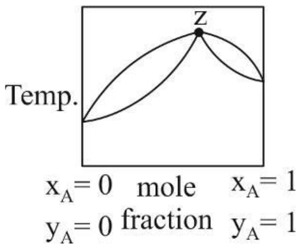
1. Elevation of boiling point of 1 molar aqueous glucose solution (Molecular weight of glucose = 180) (density = 1.2 g/ml) is:

(A) K_b **(B)** 1.20 K_b **(C)** 1.02 K_b **(D)** 0.98 K_b

2. What will be the molecular weight of $CaCl_2$ determined in its aq. solution experimentally from depression of freezing point?

(A) 111 **(B)** <111 **(C)** >111 **(D)** Data insufficient

3. A liquid mixture having composition corresponding to point z in the figure shown is subjected to distillation at constant pressure. Which of the following statement is correct about the process? (here x represents mole fraction in liquid phase and y represents mole fraction in vapour phase)



(A) The composition of distillate differs from the mixture
(B) The boiling point goes on changing
(C) The mixture has highest vapour pressure than for any other composition
(D) It represent azeotropic composition and it will not change on distillation at constant pressure

4. For an ideal binary liquid solution with $P_A^o > P_B^o$, which relation between X_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct?

(A) $Y_A < Y_B$ **(B)** $X_A > X_B$ **(C)** $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$ **(D)** $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$

5. A very dilute saturated solution of a sparingly soluble salt AB_2 has a vapour pressure of 20 mm of Hg at temperature T, while pure water exerts a pressure of 20.0108 mm Hg at the same temperature. Calculate the solubility product constant of AB_2 at the same temperature.

(A) 10^{-6} **(B)** 10^{-4} **(C)** 4×10^{-6} **(D)** 4×10^{-4}

6. The freezing point depression of a 0.10 M solution of formic acid is –0.2046°C. What is the equilibrium constant for the reaction at 298 K?

$$HCOO^-(aq) + H_2O(l) \rightleftharpoons HCOOH(aq) + OH^-(aq)$$

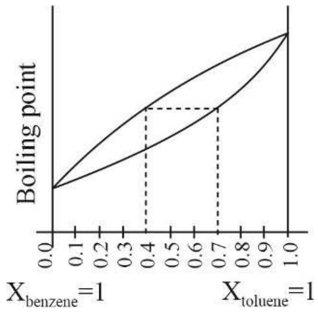
(Given: $K_f(H_2O) = 1.86 K\,kg\,mol^{-1}$, Molarity = molality)

(A) 1.1×10^{-3} **(B)** 9×10^{-12} **(C)** 9×10^{-13} **(D)** 1.1×10^{-11}

7. The solubility of common salt is 36.0 gm in 100 gm of water at 20°C. If systems I, II and III contain 20.0, 18.0 and 15.0 g of the salt added to 50.0 gm of water in each case, the vapour pressures would be in the order.

(A) I < II < III **(B)** I > II > III **(C)** I = II > III **(D)** I = II < III

8. The following graph represents variation of boiling point with composition of liquid and vapours of binary liquid mixture. The graph is plotted at constant pressure.
Which of the following statement(s) is incorrect? Here X &Y stands for mole fraction in liquid and vapour phase respectively.



(A) $X_{benzene} = 0.7$ and $Y_{toluene} = 0.4$ **(B)** $X_{toluene} = 0.3$ and $Y_{benzene} > 0.7$

(C) $X_{benzene} = 0.3$ and $Y_{toluene} = 0.4$ **(D)** if $X_{benzene} = 0.7$ then $Y_{toluene} < 0.3$

9. How many grams of sucrose (mol. Wt. = 342) should be dissolved in 100 gm water in order to produce a solution with 105°C difference between the freezing point & boiling point temperatures? (Use $k_f = 1.86\,K.kg\,mol^{-1}$; $k_b = 0.51\,K.kg\,mol^{-1}$)

(A) 34 gm **(B)** 46 gm **(C)** 72 gm **(D)** 342 gm

10. For a liquid normal boiling point is –173°C then at 2 atm pressure it's boiling point should be nearly
($\Delta H_{vap.}$ = 200 cal/mole, R = 2 cal/mol-K, ln 2 = 0.7)

(A) –73°C **(B)** 333°C **(C)** 60°C **(D)** 103°C

11. Which of the following is incorrect for a non-ideal solution of liquids A and B, showing negative deviation?

(A) $\Delta H_{mix} = -ve$ **(B)** $\Delta V_{mix} = -ve$ **(C)** $\Delta S_{mix} = -ve$ **(D)** $\Delta G_{mix} = -ve$

12. A cylinder fitted with a movable piston contains liquid water in equilibrium with water vapour at 25°C. Which of the following operation results in a decrease in the equilibrium vapour pressure at 25°C?

(A) Moving the piston downward a short distance
(B) Removing a small amount of vapour
(C) Removing a small amount of liquid water
(D) Dissolving some salt in the water.

13. Insulin is dissolved in a suitable solvent & the osmotic pressure (p in atm) of solutions of various concentration 'C' in gm/cc is measured at 27°C. The slope of the plot p against 'C' is found to be 5×10^{-3}. The molecular weight of insulin is:

(A) 4.9×10^5 **(B)** 4.9×10^3 **(C)** 4.9×10^6 **(D)** 4.9×10^4

14. In diamond, carbon atom occupy FCC lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is:

(A) 77.07 pm **(B)** 154.14 pm **(C)** 251.7 pm **(D)** 89 pm

15. $r_{Na^+} = 95\,pm$ and $r_{Cl^-} = 181\,pm$ in NaCl (rock salt) structure. What is the shortest distance between Na^+ ions?

(A) 778.3 pm **(B)** 276 pm **(C)** 195.7 pm **(D)** 390.3 pm

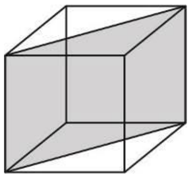
16. Which of the following will show Schottky defect:

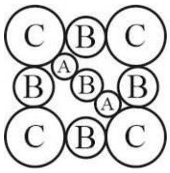
(A) CaF_2 **(B)** ZnS **(C)** AgCl **(D)** CsCl

17. In FCC unit cell, what fraction of edge is not covered by atoms?

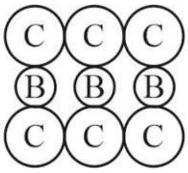
(A) 0.134 **(B)** 0.293 **(C)** 0.26 **(D)** 0.32

18. In a hypothetical solid 'C' atoms are found to form cubical close packed lattice. 'A' atoms occupy all tetrahedral voids & 'B' atoms occupy all octahedral voids. 'A' and 'B' atoms are of appropriate size i.e., there is no distortion in ccp lattice of C atoms. Now if a plane as shown in the following figure is cut, then the cross section of this plane will look like.

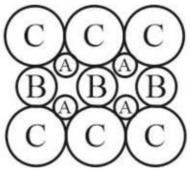




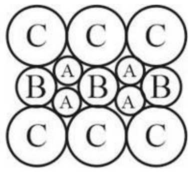
(A)



(B)



(C)



(D)

19. A solid element (monoatomic) exist as cubic crystal. If its atomic radius is 1.0 Å and the ratio of packing fraction and density is 0.1 cm^3/gm, then the atomic mass of the element is $(N_A = 6 \times 10^{23})$

(A) 8π **(B)** 16π **(C)** 80π **(D)** 4π

20. Packing fraction in 2-D hexagonal arrangement is:

(A) $\frac{\pi}{3\sqrt{2}}$ **(B)** $\frac{\pi}{3\sqrt{3}}$ **(C)** $\frac{\pi}{2\sqrt{3}}$ **(D)** $\frac{\pi}{6}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

21. A non-stoichiometric compound Fe_7S_8 consist of iron in both Fe^{+2} and Fe^{+3} form and sulphur is present as sulphide ions. Calculate cation vacancies as a percentage of Fe^{+2} initially present in the sample. [Express your answer as $\frac{100}{x}\%$ & fill x in the OMR sheet]

22. An element X (Atomic mass = 125 g/mole) crystallises in simple cubic structure. If diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. Find the density of pure element X in gm/cm^3. **[Given:** $\sqrt{3} = 1.732$, $N_A = 6 \times 10^{23}$] [Fill your answer by multiplying it with 300.]

23. If $250\ cm^3$ of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K, the ratio of the molecular masses of A and B is ______$\times 10^{-2}$ (to the nearest integer).

24. The vapour pressure of an aqueous solution is found to be 750 torr at certain temperature 'T'. If 'T' is the temperature at which pure water boils under atmospheric pressure and same solution show elevation in boiling point $\Delta T_b = 1.04\,K$, find the atmospheric pressure in torr $(K_b = 0.52\,K\,kg\,mol^{-1})$

25. At 300K, the vapour pressure of an ideal solution containing 3mole of A and 2 mole of B is 600 torr. At the same temperature, if 1.5 mole of A & 0.5 mole of C (non-volatile) are added to this solution the vapour pressure of solution increases by 30 torr. What is the value of P_B^o in torr ?

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The range of the function $f(x) = \dfrac{1}{\sqrt{x^2 - 4x + 5} - 2}$ is:

(A) $(-\infty, -1]$ **(B)** $[-1, 0)$

(C) $(-\infty, -1] \cup [1, \infty)$ **(D)** $(-\infty, -1] \cup (0, \infty)$

2. The acute angle of intersection of the curves $x^2 y = 1$ & $y = x^2$ in the first quadrant is θ, then $\tan\theta$ is the root of the equation:

(A) $3x^2 + x - 4 = 0$ **(B)** $3x^2 + 7x + 4 = 0$

(C) $3x^2 - x - 4 = 0$ **(D)** $4x^2 - 7x + 3 = 0$

3. If 't' belongs to the domain of the function $f(x) = \log_3(1 - \log_6(x^2 - 7x + 16))$, then $[t]$ cannot be: (where [.] denotes greatest integer function)

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

4. The value of $\lim\limits_{x \to -\infty} \dfrac{x^2 \tan\left(\dfrac{2}{x}\right)}{\sqrt{16x^2 - x + 1}}$ is equal to:

(A) $-\dfrac{1}{2}$ **(B)** -2 **(C)** $\dfrac{1}{2}$ **(D)** Does not exist

5. Consider the function $f(x) = \min\{|x^2 - 4|, |x^2 - 1|\}$, then:

(A) $f(x)$ has 4 points of local minima & 2 points of local maxima

(B) $f(x)$ has 4 points of local minima & 3 points of local maxima

(C) $f(x)$ has only 6 critical points

(D) $f(x)$ has 7 points of non-differentiability

6. The function $f(x) = 2x^3 - 3(a+b)x^2 + 6abx$ has a local maximum at $x = a$, if:

(A) $a > b$ **(B)** $a < b$ **(C)** $a > 0$ **(D)** $a < 0$

7. Let $f(x) = |x|$ & $g(x) = [x]$, (where [.] denotes the greatest integer function), then:

(A) $(f.g)'(0) = 0$ **(B)** $(fog)'(0) = 0$ **(C)** $(gof)'(0) = 0$ **(D)** $(f+g)'(0^-) = -1$

8. Number of integers in the range of the function $f(x) = x^2 \ln(x)$ for $x \in (0, e]$ is:

(A) 6 **(B)** 7 **(C)** 8 **(D)** 9

9. If the function $f(x)=\begin{cases} a\sqrt{x+7} & 0\le x<2 \\ bx+5 & x\ge 2\end{cases}$ is differentiable $\forall\ x\ge 0$, then (a, b) lies on:

(A) $2x+4y=5$ **(B)** $2x-4y=5$ **(C)** $4x+8y=5$ **(D)** None of these

10. Consider the function $f(x)=x^{1/x}$, $\forall\ x\in(e,\infty)$. If $g(x)$ is the inverse of $f(x)$, then the value of $g'(2^{1/4})$ is:

(A) $\frac{2^{3/2}}{1-\ln 2}$ **(B)** $\frac{4^{7/4}}{1-2\ln 2}$ **(C)** $\frac{2^{31/4}}{1-4\ln 2}$ **(D)** $\frac{2^{15/4}}{1-4\ln 2}$

11. $\lim\limits_{x\to 0}\frac{\tan^3 x-\tan x^3}{x^5}=$

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

12. Let the function $f(x)=|x+1|$. The number of values of $x\in(-2, 2)$, for which $f(x-3)$, $f(x-1)$ & $f(x+1)$ are in A.P. is:

(A) 0 **(B)** 1 **(C)** 2 **(D)** infinite

13. Which of the following functions have graphs symmetrical about origin:

(A) $f(x)=2^x+2^{-x}$ **(B)** $f(x)=\left(\ln\left(x+\sqrt{1+x^2}\right)\right)^2$

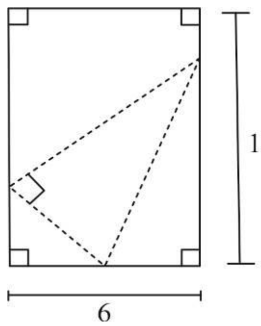
(C) $f(x+y)=f(x)+f(y)\ \forall\ x, y\in R$ **(D)** $f(x+y)=f(x)\times f(y)\ \forall\ x, y\in R$

14. If $f(x)$ is a differentiable function satisfying $|f'(x)|\le 2\ \forall\ x\in[0, 4]$ & $f(0)=0$, then:

(A) $f(x)=18$ has no solution in $[0, 4]$ **(B)** $f(x)=18$ has more than 2 solutions in $[0, 4]$

(C) $f(x)=20$ has 3 solutions in $[0, 4]$ **(D)** $f(x)=14$ has 2 solutions in $[0, 4]$

15. The lower corner of a leaf in a book is folded over so as to reach the inner edge of the page (as shown). The length of crease if the area of the folded part is minimum is:



(A) 6 **(B)** 8 **(C)** 9 **(D)** $\frac{2}{3}\sqrt{136}$

16. The shortest distance between the line $x+y+2=0$ & the curve $y=x^2$ is:

(A) $\frac{7\sqrt{2}}{4}$ **(B)** $\sqrt{2}$ **(C)** $\frac{7\sqrt{2}}{8}$ **(D)** None of these

17. Let $f(x+y) = f(x)\cdot f(y)\ \forall\ x, y \in R$ and $f(x) = 1 + x\, g(x) \ln 3$. If $\lim\limits_{x\to 0} g(x) = 1$, then $g(1) =$

(A) $\dfrac{2}{\ln(3)}$ **(B)** $\dfrac{3}{\ln(3)}$ **(C)** $\dfrac{\ln(3)}{2}$ **(D)** None of these

18. Let $f(x) = 8x^3 + 4x^2 + 2kx + 1$ where $k \in R$ if $f(x)$ is strictly increasing in *R*, then range of values of *k* is:

(A) $\left(-\infty, \dfrac{1}{3}\right]$ **(B)** $\left[\dfrac{1}{3}, \infty\right)$ **(C)** $\left[\dfrac{1}{3}, 1\right)$ **(D)** $\left[-\dfrac{1}{3}, \infty\right)$

19. If $y = at^2 + 2bt + c,\ t = ax^2 + 2bx + c$, then $\dfrac{d^3y}{dx^3} =$

(A) $24a^2(at+b)$ **(B)** $24a(ax+b)^2$ **(C)** $24a(at+b)^2$ **(D)** $24a^2(ax+b)$

20. $\lim\limits_{x\to 0}\left[\dfrac{3\sin x}{x}\right] + \left[\dfrac{4x}{\tan x}\right]$ is ([.] denotes greatest integer function)

(A) 5 **(B)** 6 **(C)** 7 **(D)** Does not exist

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25, 0.08)**

21. Smallest positive integer in the range of function $f(x) = (x+2)e^{\ln(x+2)} + \dfrac{e^{-\frac{1}{\log_x e}}}{x} - \dfrac{2x}{e^{-\ln 2}}$ is _______.

22. The natural number n for which $\lim\limits_{x \to 0} \dfrac{(27^x - 9^x - 3^x + 1)(\cos x - e^x)}{x^{2n+1}}$ is a finite non zero number, then n is ______.

23. The left hand derivative of $f(x) = [x]\sin(\pi x)$ at $x = -5$ is $k\pi$, then k is_____. ([.] denotes greatest integer function)

24. Let $f(x)$ be an increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$, the number of integers in the range of a is ________.

25. If $f(x) = 4x^3 - x^2 - 2x + 1$ & $g(x) = \begin{cases} \min\{f(t) : 0 \le t \le x\} & 0 \le x \le 1 \\ 3 - x & 1 < x \le 2 \end{cases}$

, then $2\left(g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)\right)$ is ______.

SPACE FOR ROUGH WORK

End of JEE Main - 10 | JEE 2022